Heterogeneous response and spillover effects of SSB taxes *

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Abstract

In this paper we evaluate the taxation of sugar-sweetened and edulcorated beverages by using the natural quasi-experiment in Spain: in 2021 an increase in Value-Added Tax was introduced in all regions of Spain except for some of them, which do not have VAT. These regions serve as a control group offering a unique opportunity in the literature because, unlike most previous work that uses sales data, we use micro-data from a rich survey of households with information on their expenditures and quantities purchased. We find a price tax pass-through of more than 90% and a 12% fall in soft drink consumption among the poorest tertile of households, especially among those with children aged 5-16. Novelly, we find an almost equivalent reduction in expenditure on complementary, unhealthy goods among the same households in the first tertile. However, the remaining households did not react to the tax by reducing either their consumption of soft drinks or their consumption of complementary goods. Our results show the importance of considering the household structure as well as the response of the consumption of complementary goods when assessing the effect of this type of tax on consumption.

Keywords: sugar-sweetened beverages, tax, consumption, complementary

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1 Introduction

Taxation to correct negative externalities in consumption has a long tradition in economic history. However, it is only in the last decade that such taxes have started to be levied on sugar-sweetened beverages, mostly in high-income countries. Although the empirical literature analysing this type of tax is recent, a good number of studies have already begun to accumulate around three key issues of this policy: the pass-through of the tax to prices, its effect on consumption and its distributional effect. The results so far have been very heterogeneous. For example, in the literature review carried out by Cawley et al. (2019), the range of pass-through of these taxes to final prices varies between 43% and 100%, depending on the case. In the case of demand elasticities, Andreyeva et al. (2010) estimates, based on fourteen studies, an average aggregate elasticity of -0.74 and a range of variation from -0.13 to -3.18. Meanwhile, in another meta-analysis, Powell et al. (2013) obtains higher results of the order of -1.21. Partly because of the heterogeneity of results in the literature, the design of this type of tax to correct the negative externalities of sugar-sweetened beverages is still an open debate. On the one hand, authors such as Sharma et al. (2014) favour volumetric taxes, both for efficiency and equity reasons, although these are not without problems, including potential demand-side substitution effects, as will be discussed in more detail in the literature review.

In January 2021, an increase in Value Added Tax (VAT) on sugar-sweetened and edulcorated beverages from the reduced rate of 10% to the general rate of 21% came into effect. This measure has two particular features. Firstly, it does not apply only to sugar-sweetened beverages, *zero/ligth* products were also affected by the policy as edulcorated beverages were also included. Secondly, sugar-sweetened or edulcorated beverages consumed in bars or restaurants were excluded from the increase. This situation allows us to study the effect of taxation on the negative externalities of sugar-sweetened beverage consumption, addressing a growing concern in the literature: the substitution effect between sugar-sweetened and edulcorated beverages. For example, in the case of Catalonia, Castelló and Casasnovas (2020) identifies a significant substitution effect between sugar-sweetened and edulcorated beverages, which experienced different price increases as a consequence of the tax design. By applying the VAT increase across the

full range of sugar-sweetened or edulcorated beverages, this policy provides a new case study of the consequences of expanding the list of goods on which the tax is applied.

Our empirical strategy uses the fact that certain regions such as the Canary Islands and the autonomous cities of Ceuta and Melilla have their own indirect taxes: the Canary Island General Indirect Tax (IGIC) and the Tax on Production, Services and Imports (IPSI) respectively, so that households in these territories did not experience any increase in the tax burden on sugar-sweetened and edulcorated beverages. Using these three regions as a control group, and thanks to the short notice of the announcement of the measure (late 2020), we establish the causal impact of the tax on prices, consumption and household spending. To conduct the analysis, we use micro-data from the Household Budget Survey (HBS) for the years 2017 to 2021, a rich annual household survey that provides detailed information on a representative sample of around 20,000 Spanish households, including their spending on a large number of groups of goods and services, which allows us to identify very clearly the effect of the tax on different categories of affected products.

First, we find that the degree of pass-through to prices of the VAT increase amounts to 92%, corresponding to an increase in the price per litre of soft drink of 11 cents, an increase of 9.2%. Using data corresponding to the Consumer Price Index (CPI), we reach an almost identical conclusion, with a pass-through of close to 100%, concentrated in the month of January. In other words, the pass-through of the tax into prices was almost complete and occurred very quickly. We made additional estimates for other smaller groups of sugar-sweetened and edulcorated beverages, finding a degree of pass-through of slightly more than 100%, consistent with the findings of Bonnet and Réquillart (2013) for France.

On the other hand, we find that the effect of the tax on the amount of soft drinks consumed is very heterogeneous according to the household's economic capacity. For households in the first tercile of equivalent expenditure, we find a negative and highly significant effect amounting to 10.8 litres per household, which is equivalent to a 12% drop in consumption. For the second tercile the result is negative, but not statistically significant. For the third tercile, the coefficient is positive, but again not significant. It should be noted that the results for the third tercile should be taken with caution as

our control regions are particularly poor, which, coupled with their limited sample size, means that the representativeness of households in the upper part of the distribution is compromised. Additional estimates allow us to decompose the fall in consumption of households in the first tercile, as a consequence of the policy, into a fall via the intensive and extensive margin. We find that 79% of the fall in the quantity of soft drinks consumed is explained by a fall in consumption by households with positive consumption, which account for 60% of the total.

In addition, we contribute to the literature by studying the heterogeneous effect of the tax by household composition, focusing on the differential effect of the tax on households with children of age to consume soft drinks. Our results indicate that, among households with lower economic capacity (first tercile), the effect of the consumption tax was several times larger for households with children than for households without children. Specifically, we document a drop in consumption among poor households with children of 24.6 litres of soft drinks per household per year, compared to an average drop of 10.8 litres in the first tercile.

Finally, our paper studies an issue so far underrepresented in the literature: the possible spillover effects of these taxes on the consumption of complementary goods. To do so, we replicate our empirical strategy with the expenditure on soft drinks of the poorest households who, as we have documented above, are the only ones to reduce their consumption of soft drinks as a consequence of the tax. We find that these households reduced their expenditure on snacks by almost the same proportion as they reduced their consumption of soft drinks, showing a fall of 11.5%, a novel finding that highlights the importance of taking into account complementary goods in the impact assessment of taxes on sugary products.

The rest of the paper is structured as follows: the next section provides a review of the literature, followed by an explanation of the institutional environment in which the policy is developed, the methodology and the databases to be used, the results are presented and discussed, and the paper ends with a section devoted to the main conclusions.

2 Contribution to the literature

The extent to which a tax changes the consumption habits of households has been extensively studied in the academic literature. Thus, Arce (2022) studied the impact of the increase in reduced VAT in 2012 on Spanish household spending. This article finds a reduction in average household spending, especially on non-alcoholic beverages, restaurants and hotels.

In recent years, levies on sugar-sweetened beverages have been introduced around the world, highlighting that the optimal tax burden will be conditioned by the level of externalities, internalities and regressivity. However, a broader review of the existing literature is essential to understand the different results that have been found across the various evaluations, especially what effects such taxes have had on prices, sales and consumer health. Although these taxes have different designs and implementations, the empirical evidence shows an increase in the average price per litre and a negative effect on the consumption of these beverages.

France started with the taxation of sugar-sweetened beverages in January 2012 by establishing a tax of 0.0716 € per litre of beverage, excluding VAT or 0.0755€ including VAT. This tax affects soft drinks, diet soft drinks and juices with added sugar, excluding pure juices and water. Two years after the implementation of this tax, (Capacci et al., 2019) evaluated the effects of the taxed goods on prices and consumption. They compared, using the difference-in-differences method, the level of prices and consumption between two neighbouring regions in France (treatment) and Italy (control), before and after the French tax came into force. To carry out their study they used monthly Consumer Price Index data for both countries, micro data from the Household Budget Survey for the two Italian regions and Kantar's household consumption panel for the French regions. Thus, (Capacci et al., 2019) found that the tax had been passed on to prices by 66% on average, leading to a reduction in consumption of sugar-sweetened beverages of about half a litre per person per year.

In 2017, Catalonia began to tax drinks with between 5 and 8 grams of sugar per 100ml with 0.08 €/L, and those drinks with more than 8 grams/100ml of sugar with

0.12 €/L, forcing establishments to pass on 100% of the tax in the final price. Recently, (Castelló and Casasnovas, 2020) and (Royo-Bordonada et al., 2022), evaluated the impact of this tax on consumption, finding a drop of 7.7% and 16.7% respectively. Since Zero/Light drinks did not suffer from this tax burden, they quantified an increase in consumption of 21.7% in this type of drinks, thus finding a substitution effect.

A very similar tax to that of Catalonia was introduced in Portugal in 2017, where a tax of 0.08 €/L was levied on sugar-sweetened and edulcorated beverages with less than 80 grams of sugar per litre and 0.16 euro/L on soft drinks with more than 80 grams per litre. Thus, they found that the tax caused a price increase equivalent to 100% of the tax increase and a reduction in consumption of 18%.

To combat overweight, Mexico introduced a 1 peso/litre tax in January 2014 on beverages with added sugar. In the study by (Grogger, 2017) they analyse the effect this tax had on the prices of these products, using the Consumer Price Index. This study finds an increase in the prices of taxed products of 1.32 pesos over the course of 2014.

Several cities in the United States have imposed taxes on sugar-sweetened beverages in recent years. In January 2015, the city of Berkeley, California, was the first to introduce a 1 cent per ounce tax on soda. In a study, using sales data from all Berkeley supermarkets and a sample of San Francisco supermarkets, (Cawley and Frisvold, 2015) found, on average, a 43.1% price increase. However, for every extra mile of distance from San Francisco, the price increased by between 25.8% and 33.3

Subsequently, in 2017 the city of Philadelphia taxed soft drinks with 1.5 cents per ounce. Where (Cawley et al., 2019) found a negative effect on consumption of 8.9 ounces per purchases, while increasing purchases by out-of-town residents of the tax. On the other hand, (Seiler et al., 2021) also for Philadelphia, they estimated a 97% pass-through of the tax to final prices and a 22% drop in consumption of the taxed beverages.

Consumption of sugary drinks increases the likelihood of being overweight and experiencing cardiovascular disease, diabetes, etc. Malik et al. (2013). The effect of

sugar-sweetened beverage taxes on the body mass index of young people in Mauritius has been studied, finding only a negative effect on the likelihood of children consuming sugar-sweetened beverages. In the same way, they find no solid evidence that existing taxes reduce adult body weight and open the possibility that higher levels of taxation are needed to achieve this goal.

Table 1: Resumen de la literatura existente

Articulo	Región	Tax	Incluye diet drinks ?	Diseño	Precios	Consumo
Castelló and Casasnovas, 2020	Cataluña 2017	5-<8 gr/100ml : 0.08 euros >8 gr/100ml : 0.12 euros	No	DiD	100%	-7.7%
Royo-Bordonada et al., 2022				DiD	100%	-16.7%
Capacci et al., 2019	Francia, 2012	0.0716 euros por Litro	Yes	DiD	66%	0.5L per capita.
Gonçalves & Dos Santos, 2020	Portugal, 2017	<80 gr/L : 0.08 euros >80 gr/L : 0.16 euros	Yes	DiD	100%	-18%
Grogger, 2017	Mexico, 2014	1 peso por litro	No	DiD	132%	-
Cawley & Frisvold, 2015	Berkeley, 2015	1 centavo por onza	No	DiD	43.1%	-
Cawley et al, 2019	Philadelphia, 2017	1.5 centavos por onza	Yes	DiD	-	-8.9 onzas por compra
Seiler et al., 2021				DiD	97%	-22%

Note: Elaboración propia de los autores

3 Data and Institutional setting

3.1 The data

The analysis is based on microdata from the Household Budget Survey (HBS), an annual survey published by the Spanish National Statistics Institute (INE), which collects very detailed information on the expenditure of a representative sample of Spanish households. The HBS also provides information on the composition of the household, its characteristics, members, type of dwelling and tenure status. Additionally, it includes characteristics of the main breadwinner such as employment, country of birth, nationality and age.

With respect to the different expenditure groups, we will focus on the analysis of the group corresponding to code 01222 "Carbonated and non-carbonated soft drinks", which is the largest group of goods, related to sugar-sweetened and edulcorated beverages, in terms of expenditure and which contains a higher percentage of goods affected by the VAT increase. However, in the annex we study the effect of broadening the definition of sugar-sweetened and edulcorated beverages by including other minor expenditure codes such as energy drinks and isotonic drinks, which do not alter the main conclusions of the paper.

Using the HBS for the analysis offers several advantages over other databases. Firstly, its breadth in terms of sample and household information allows different heterogeneity analyses to be carried out. In addition, the HBS offers the possibility of obtaining the three key variables for the analysis, expenditure on soft drinks, consumption in litres and price, all at household level. Although the ultimate purpose of the survey is to measure total household expenditure, the amount consumed is collected in many expenditure categories, although in some cases this variable has problems of missing values. Fortunately for us, the percentage of households that report some expenditure on soft drinks, but have missing values for their consumption in litres, is only 3% of the total number of households in the whole period analysed. The average expenditure on soft drinks of this 3% of households is barely 2 €, and is evenly distributed by terciles of equivalent expenditure, so it does not threaten the robustness of our results

in any way. However, the HBS does not have income data from administrative records, since the income variables in the survey are self-reported by the household. Therefore, throughout the analysis we will use the equivalent household expenditure as a proxy for the household's economic status. We define equivalent household expenditure as total household expenditure divided by the number of consumption units (using the modified OECD scale) that make up the household, following the definition used by the INE.

Table 2: Number of households by period and tercile of equivalent expenditure

	Tercil 1		Tercil 2		Tercil 3	
	Tratados	Controles	Tratados	Controles	Tratados	Controles
2017	6.783	652	7.076	327	6.942	263
2018	6.627	625	6.762	368	6.770	243
2019	6.439	569	6.675	341	6.554	239
2020	5.684	532	6.158	321	6.231	243
2021	5.849	530	6.315	312	6.115	273
Total	31.382	2.908	32.986	1.669	32.612	1.261

*S*ource: Own elaboration based on microdata from HBS (2017-2021)

Information is available for 103,000 households over a five-year period, of which approximately 60% have positive soft drink consumption. Of the total households available, about 5% of them belong to the control group, with most of them located in the Canary Islands region. Since the households belonging to our control group are at the lower end of the equivalent expenditure distribution, as shown in the table 2, we will present the results disaggregated by terciles of equivalent expenditure. The reason for using terciles of equivalised expenditure, rather than alternative measures such as quartiles or quintiles, is precisely to ensure the representativeness of our control group at the top of the distribution. The portion of this subgroup of control households in the top 33% of equivalent expenditure that has positive expenditure on the main group of sugar-sweetened and edulcorated beverages is in line with that of the sample as a whole: around 60%.

In addition to the Household Budget Survey (HBS), we have data, aggregated by month, from the Consumer Price Index (CPI) to complement the price analysis. These data are published monthly by the INE and allow us to check, firstly, whether there has been any type of anticipation effect on prices and, secondly, in which part of the

year 2021 the reaction of prices, on the part of companies, to the growth of the tax was concentrated.

3.2 Institutional Context

Value Added Tax (VAT) is the indirect tax with the highest revenue in Spain. As in most European countries, it has different tax brackets for different goods and services. In the case of Spain, there are four rates: the general rate of 21%, the reduced rate of 10%, the super-reduced rate of 4% and certain goods and services which, being exempt, are taxed at 0%. Spain is one of the European countries with the highest percentage of the VAT tax base being taxed at reduced rates (Airef, 2020), which is where sugar-sweetened and edulcorated beverages were until 2021: they were taxed at the reduced rate of 10%.

On 27 October 2020, the Spanish government published its draft General State Budget (PGE), which included a proposal to move these drinks from the reduced rate to the general rate, which meant an increase of 11 percentage points in the tax rate. The measure excluded sugar-sweetened and edulcorated beverages consumed in bars and restaurants, so as not to hinder the recovery of the sector after the Covid-19 pandemic. Although the government initially quantified the expected revenue at 340 million euros, the Independent Authority for Fiscal Responsibility (AIReF) (Airef, 2020)lowered this forecast to 208 million euros, and later to 189 million euros, after excluding liquid yoghurts from the policy. It is worth noting that AIReF used the same Household Budget Survey used in this paper for its estimates, although no household consumption feedback was incorporated in their work.

Following the favourable vote in Congress on 3 December, the tax hike included in the SGB came into effect on 1 January 2021, thus leaving two months for companies and consumers to react to the policy in advance. In the following points we will provide evidence that discourages from thinking that this anticipation effect has taken place, based on price data from the CPI. The companies in the sector, grouped around the Spanish Federation of Food and Drink Industries (FIAB), expressed their disagreement with the policy in two statements, saying that "sugar taxes do not change consumption

habits or have an impact on health" and estimating the effect of the tax at a loss of between 2,000 and 6,000 jobs. [1]

4 Empirical Strategy

As mentioned above, our control group are the households located in the Canary Islands, Ceuta and Melilla, while the treatment households are those located in the Spanish mainland or in the Balearic Islands. Our empirical strategy consists of a difference-in-differences model, using the cross-sectional data from the HBS. Our main specification is the one described in the equation 1, which we will apply to three different independent variables, prices, consumption in litres and expenditure in euros on soft drinks for each household.

$$Y_{itc} = \alpha + \beta_1 Treat_i * Post_t + \lambda_t + \delta_c + \theta_i + \epsilon_{it} \tag{1}$$

Where β_1 is the coefficient of interest associated with the policy effect, Y_{itc} is the variable of interest, whether it be price, quantity consumed or expenditure on soft drinks of household i in period t and region c. In turn, λ_t represents the time fixed effects and δ_c represents the region fixed effects, and finally θ_i is a set of controls for observable household characteristics such as the age of the main breadwinner, educational attainment, activity category and gender, as well as other associated household controls such as the size of the municipality of residence, the number of members and the total equivalent expenditure of the household.

In addition, it is of interest to know whether the effect of the policy has been different across households with children, so we estimate an additional regression derived from the main regression to elucidate this. To do so, we introduce an interaction term, as can be seen in the equation 2 in which the effect of the policy is interacted with a binary variable that takes zero if there are children between 5 and 16 in the household and one otherwise, associated with β_3. In this way the coefficient β_1 now gives us the

[1] https://fiab.es/fiab-rechaza-la-subida-del-iva-a-las-bebidas-azucaradas-y-edulcoradas/

effect of the policy on households with children in that age group, since they are the group that has been left as a reference in the regression.

$$Y_{itc} = \alpha + \beta_1 Treat_i * Post_t + \beta_2 Treat_i * Post_t * Niños_i + \beta_3 Niños_i + \lambda_t + \delta_c + \theta_i + \epsilon_{it} \quad (2)$$

Our empirical strategy requires two essential assumptions. First, that people from the treated regions do not travel to the control territories to make their purchases. This does not seem likely, since our control regions are located outside the peninsula and travel costs would be too high. Secondly, the fundamental assumption, as in any difference-in-differences analysis, can be expressed as $E[\epsilon_{it}|Treat_j * Post_t = 0$, i.e., that the assumption of parallel prior trends is met. In addition to assessing that these trends are met, by testing parallel prior trends hypotheses, we estimate additional lead and lag regressions to confirm this assumption, where 2020, the year prior to policy implementation, will serve as the base year.

$$Y_{itc} = \alpha + \beta_t Treat_i * \lambda_t + \lambda_t + \delta_c + \theta_i + \epsilon_{it} \quad (3)$$

For the reasons explained in section 3, the above regressions will be run separately for each of the three terciles of equivalised expenditure. These terciles are defined annually, using the entire available sample of households. The equivalent expenditure formula uses the total monetary expenditure of the household divided by the number of consumption units (modified OECD scale) that make up the household.

5 Results

5.1 Effects on final prices

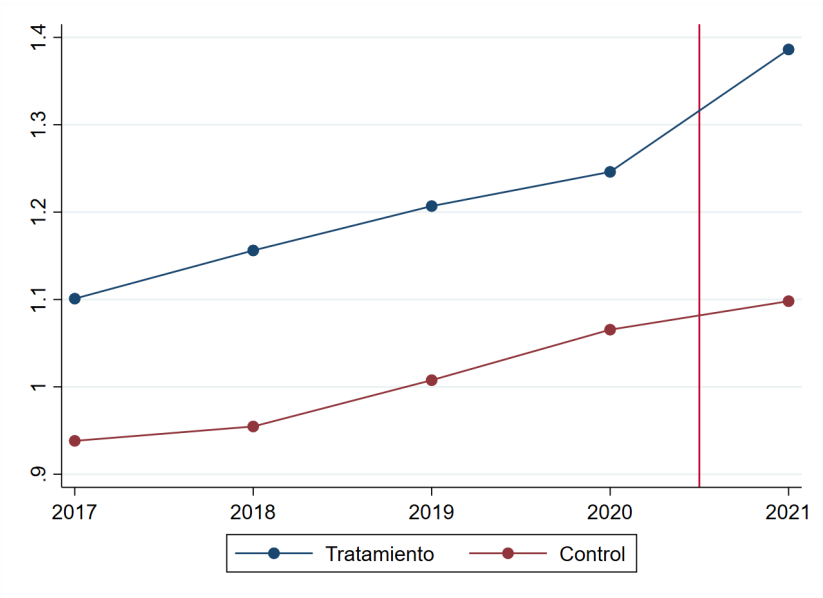
First, we begin by assessing the extent to which the tax was passed through to soft drink prices. To do so, we calculate the price per litre of soft drink from the HBS microdata, using consumption data in litres and expenditure data, which already includes taxes. In total, we have price information from 58% of households, which report consumption and expenditure information, a percentage that hardly varies between the different

terciles of equivalent expenditure.

Below, we show the average price per litre of soft drink, differentiating between households in the treatment group (Peninsula and Balearic Islands), and those in the control group (Canary Islands, Ceuta and Melilla) between 2017 and 2021. As can be seen, the difference in average price between the two groups remains constant until 2021, when the average price in the treatment group increases by 11.2%.

There is a significant difference in the average price between treatment and control group households, which can be partially explained by the difference in tax rates between VAT and IGIC. While soft drinks were taxed at 10% VAT until 2021, the tax rate associated with IGIC was only 3% between 2017 and 2021. Despite this difference in price levels, the evolution of prices in the whole period before the policy change is very similar, as can be seen in the graph 1.

Figure 1: Price per liter of soft drink in € (2017-2021)

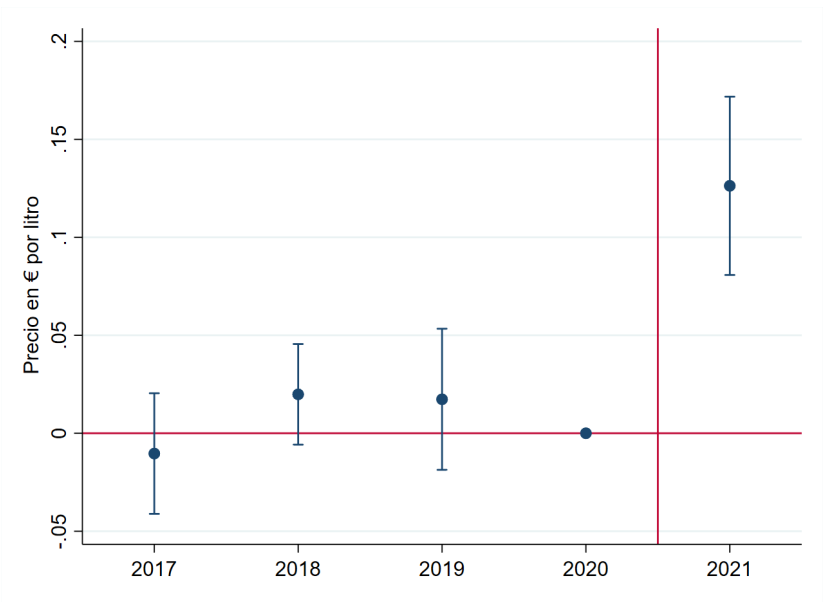


To validate our identification strategy, we estimate a leads and lags model on the price per litre of soft drinks, including autonomous community (AC) and year fixed effects, a series of basic household controls and with clustered errors at the AC level. The graph 2 shows the coefficients associated with the interaction terms, defined as the binary year variables multiplied by the treatment, leaving as a base the coefficient corresponding to the year 2020. The coefficients associated with the years 2017, 2018, and 2019 are all statistically insignificant and very close to zero. However, the coef-

ficient associated with the interaction between treatment and 2021 is positive, highly significant, and economically relevant, at 12 cents per litre.

In the table 3, we show the results of the equation 1 applied, separately, to the price per litre of soft drinks, energy drinks, and isotonic drinks. In our preferred specification, which includes household and mainstay controls, we find a final policy effect of 12 cents per litre on the price of soft drinks. The inclusion of these controls slightly increases the coefficient associated with the policy effect by one cent, suggesting little change in the treatment group relative to the control in terms of its observable characteristics after the policy. This effect is equivalent to a 9.6% increase in prices as a result of the policy compared to prices in the year before the VAT increase. Since the theoretical price increase that should occur if the tax were fully passed through is 10%, we can conclude that 96% of the tax burden was passed through to final prices. It should be recalled that, unlike the tax on sugar-sweetened beverages in Catalonia, this tax increase had no legal requirement to pass on the tax to prices, so these results are entirely attributable to the behaviour of market agents.

Figure 2: Effect of the VAT increase on the price of soft drinks



Also in table 3, we can see the estimates of the same model for two other groups of goods, isotonic drinks and energy drinks. These groups, as can be seen from the number of observations, have significantly fewer consuming households. The results, in our preferred specification, show a price increase of 0.7 €/litre for isotonic drinks and 0.5 €/litre for energy drinks. This represents a percentage increase, compared to their

price in the year before the policy, of 20% and 23% respectively. The degree of pass-through to prices is therefore well over 100%. While this may seem counterintuitive at first sight, it is consistent with greater vertical integration of firms in the sector, and more inelastic demand in these particular products, as shown in (Bonnet and Réquillart, 2013) for the case of France. In the annex we provide evidence that the assumption of parallel prior trends was also fulfilled in the case of these two products.

Table 3: Results of the price model

	(1)	(2)	(3)
Refreshing drinks	0.105***	0.118***	0.120***
	(0.020)	(0.021)	(0.022)
Observations	62,193	62,193	62,193
Energy drinks	0.797*	0.739*	0.698*
	(0.400)	(0.395)	(0.360)
Observations	7,124	7,124	7,124
Isotonic drinks	0.565***	0.592***	0.510***
	(0.152)	(0.134)	(0.118)
Observations	12,663	12,663	12,663

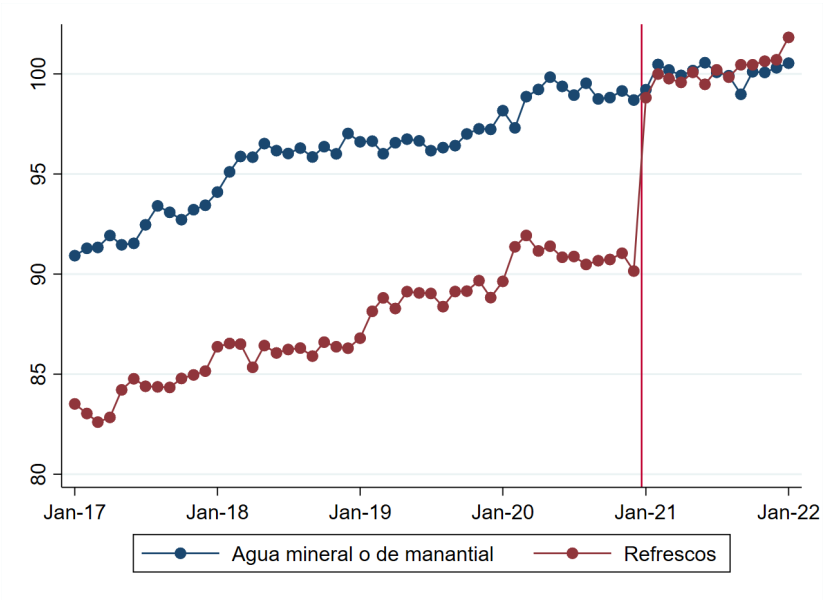
Results of the equation 1 for the prices of each good, expressed in euros per liter. Clustered standard errors at the CCAA level. SP: Main breadwinner of the household.*** p<0.01, ** p<0.05, * p<0.1

In light of these results, the time it took for companies to react and incorporate the tax increase into prices is a relevant question. On the one hand, there could be anticipatory behaviour in the two months between the announcement of the measure and its entry into force, or the pass-through to prices could be fully implemented in 2021 but take several months to be completed, both of which could affect our subsequent estimates of consumption and expenditure. We therefore consulted CPI data for soft drinks and bottled water from January 2018 to December 2021, shown in the graph below. Using bottled water as a control group, following (Capacci et al., 2019), it can be clearly seen that the price growth of both goods was in fact parallel before the policy came into force. The graph 3 answers the above questions: firstly, no price increase for soft drinks can be seen in the two months prior to the entry into force, where the anticipation effect could occur. Moreover, it can be seen that the incorporation of the tax into final prices occurs essentially in the month of January and, to a lesser extent,

February. Considering the increase in prices in both months, the degree of pass-through to prices is around 98%, barely 2 percentage points higher than our previous estimate.

The evidence presented allows us to conclude that, despite there being no legal imperative to pass the tax on to prices, companies in the sector passed on more than 90% of it to final consumer prices. For the next section, it is worth remembering that, according to the World Health Organisation (WHO), price increases of at least 20% are necessary as a consequence of these taxes to produce significant effects on calorie consumption, weight and various diseases.

Figure 3: CPI evolution between 2017 and 2022 (2021=100)



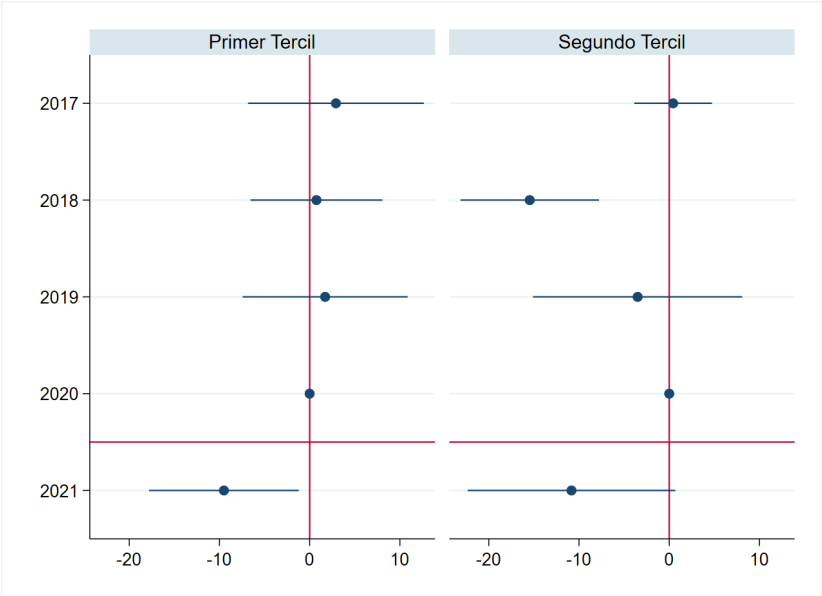
5.2 Effects on soft drink consumption

Having found in the previous section that the effect of the price tax was large and statistically significant, it is necessary to assess what effect this price increase had on household consumption of soft drinks. The specification of the difference-in-differences model is the same as that described in the equation 1, except that in this case we will provide the results by terciles of equivalent household expenditure, the variable we use as a proxy for household economic capacity. The reason for this is twofold: on the one hand, it helps us to study the heterogeneous effects of the policy, depending on the household's economic capacity. Moreover, as there are problems of representativeness of the top tercile among the control group, we believe it is convenient to take the results

of this group with caution as they fulfil less rigorously the assumption of parallel prior trends.

In the following graph we can see the leads and lags regressions for the first two terciles of expenditure. In the case of the first tercile of equivalent expenditure, all coefficients for the period before the policy change are positive, but statistically insignificant, while the coefficient for 2021 is negative, significant and equivalent to 7.5 litres. For the second tercile, the coefficient associated with 2018 is statistically significant and negative, as is the coefficient for 2021, although the coefficient for one of the years prior to the policy change is significant in the annex table 14 we perform a parallel prior trends test where the hypothesis of non-parallel prior trends is rejected with a p-value of 0.188.

Figure 4: Effect of the VAT increase on the consumption of soft drinks in liters



In the third tercile of equivalent expenditure, the same parallel prior trend contrast is rejected at 5% significance, though not at 1%, which is probably explained by the very small sample size among the control group, which dramatically increases the variability from year to year. As seen in the table 2, only 20% of households in the control group are in this tercile, which amounts to just 200 households per year, of which just over 100 would consume soft drinks. Next, we show the results of the equation 1 by terciles, using household consumption in litres as the independent variable, including those with zero consumption. Within the set of control, we add in the regressions, we include total equivalent household expenditure, to control for heterogeneity of economic status

within each tercile. In addition, we include the characteristics of the main breadwinner such as gender, country of birth, educational attainment and age. Finally, we include the number of household members and the type of household, as well as year and region fixed effects.

Table 4: Effects of the VAT increase in 2021 on the consumption of soft drinks per household (Liters)

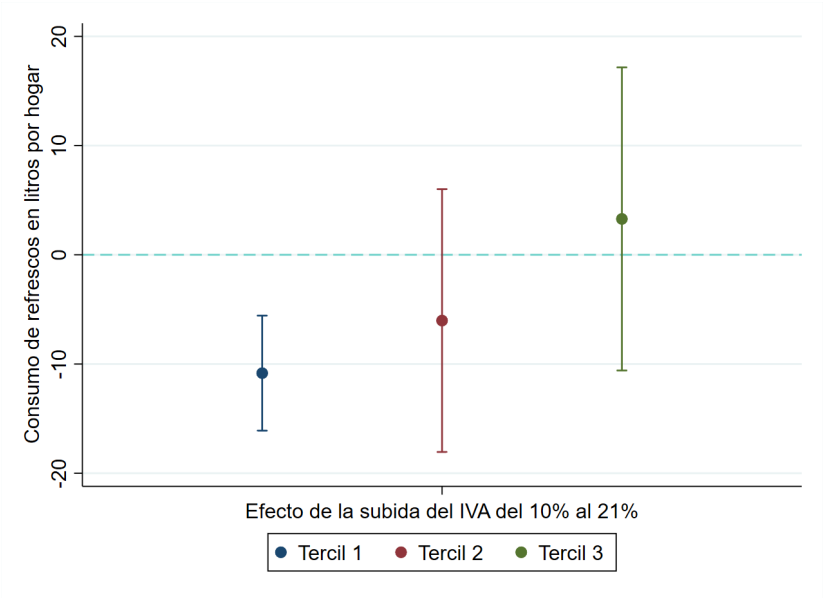
	(1)	(2)	(3)
Variables	Sin controles	Controles SP	Controles SP y hogar
All households			
Tercil 1	-11.35***	-11.52***	-10.83***
	(3.351)	(3.153)	(2.586)
Tercil 2	-3.306	-7.036	-6.022
	(5.957)	(5.628)	(5.727)
Tercil 3	10.85	5.005	3.285
	(6.748)	(6.807)	(6.605)
Observations	34,290	34,290	34,290
Only for households with soft drink consumption			
Tercil 1	-15.01***	-15.05***	-14.25***
	(4.497)	(4.769)	(4.227)
Observations	20,311	20,311	20,311
Tercil 2	-1.912	-7.843	-6.684
	(7.319)	(6.720)	(6.794)
Observations	21,362	21,362	21,362
Tercil 3	17.95*	11.73	9.691
	(8.781)	(8.703)	(8.668)
Observations	20,520	20,520	20,520

Results of the equation 1 for the consumption of soft drinks, expressed in liters per household. SP: Main breadwinner of the household. Clustered standard errors at the Autonomous Community level.*** p<0.01, ** p<0.05, * p<0.1

In the case of the poorest 33% of households, the VAT increase generated a negative and very significant fall in soft drink consumption of 10.8 litres per household per year, which is a very significant effect on their previous consumption, which amounted to 83 litres per household per year in the year before the policy was implemented. This implies a fall in consumption of 13% in response to a price increase of close to 10%, implying a price elasticity of demand for households in the first tercile of -1.3. The

second and third terciles show negative and positive effects of the policy on consumption respectively, but in neither of them are the results statistically significant at the usual significance levels.

Figure 5: Effect of the VAT increase in 2021 on the consumption of soft drinks by tertiles of equivalent expenditure



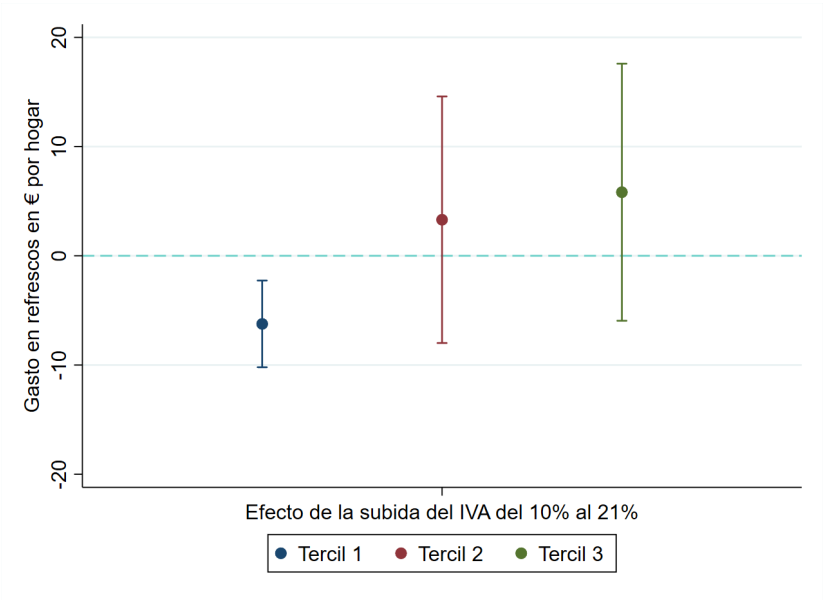
It should be noted that the above results refer to households as a whole, so that the fall in consumption observed in the first tercile can be explained either by a reduction in consumption among those households with positive consumption (intensive margin), or by an increase in the number of households that do not consume soft drinks (extensive margin). To decompose the fall in consumption among households in the first tercile between the two components, the same model is estimated only for households with positive soft drink consumption, the results of which are shown in the second part of the table 4. The fall in consumption among households in the first tercile now becomes 14.2 litres per household, considering that about 55% of households have positive soft drink consumption, that leaves a weighted fall attributable to the intensive margin of 8 litres. In other words, 80% of the negative effect associated with the policy would be caused by a lower consumption of soft drink-intensive households, while the remaining 20% could be attributed to a slight fall in soft drink-intensive households.

These results are reasonably in line with previous literature on the effect of such taxes on consumption. For example, (Castelló and Casasnovas, 2020) found a drop in sales in Catalonia of 7.7% concentrated in goods with a higher amount of sugar

and therefore with a higher tax rate, while for products with a lower sugar content, which experience a similar rise to the one studied in this article, the drop in sales is not significant. On the other hand, (Capacci et al., 2019) for France does not find an aggregate effect on household consumption but, as in our case, they do find an effect on certain groups of households, with households with higher consumption showing a clear fall in consumption as a result of the tax increase.

The effect that the policy has had on soft drink consumption in less affluent households has consequently had a negative effect on the initially expected revenue from the tax increase. Although the fall in this expected revenue was only 12% and, therefore, the tax burden has also increased among these households, it is worth noting that the final effect of the policy has been a slight fall in total household expenditure on soft drinks. To check this, we estimate the equation 1 this time with household expenditure on soft drinks, including taxes, and applying the same controls as in the previous regressions. The results are in the expected direction, with the last two terciles showing an increase in expenditure, consistent with the increase in prices, although this increase is not statistically significant. For households belonging to the first tercile, the effect of the policy is negative and significant, reducing expenditure per household by 6 €, a fall of around 6% from previous levels.

Figure 6: Effect of the VAT increase in 2021 on the expenditure on soft drinks by tertiles of equivalent expenditure



Finally, one of the main objectives of the policy is to reduce childhood obesity rates,

although we do not have medical data on household members, we can study whether the policy had a greater effect on consumption in households with children. To do so, we introduce a new variable in our main specification that interacts the effect of the policy with a binary variable that takes a value of zero if there are children between 5 and 16 years old in the household and one otherwise, thus excluding children under five years old because we consider that most of them are still too young to consume this type of beverage. The results indicate that, while in the last two terciles the interaction term is not statistically significant, it is significant in the case of the first tercile, indicating a larger effect of the policy on households with children in the age range we have defined.

Specifically, the effect of the VAT increase on soft drink consumption among households without children aged 5-16 in the first tercile falls from the previously estimated ten litres to five litres while the interaction term rises to twenty and a half litres. This indicates a drop in consumption among households with children in the first tertile to 24.6 litres per household. It should be noted that poor households with children had, before the policy change, a higher level of soft drink consumption than households in the same tertile without children. While the former reached 125 litres per household in 2019, the latter barely exceeded 70 litres. In percentage terms, the response of consumption to the policy is therefore less pronounced among both groups of households in the first tercile. Among households with children the fall would be about 20% while for poor households without children it would be about 7%. The following table shows the results of the equation 2 considering different levels of controls.

Although the size of the fall in expenditure as a result of the policy among poor households is not economically very relevant, since it pits the price increase against the fall in quantities consumed, it is worth putting the size of the fall in consumption into perspective. A fall in consumption of 10 litres represents a fall in consumption of 30 cans of soft drinks per household per year, rising to 44 for households with positive consumption and up to 75 for poor households with minors.

Table 5: Effect of the VAT in 2021 increase on households with children between the ages of 5 and 16

	(1)	(2)	(3)
Variable	Sin controles	Controles SP	Controles SP y hogar
Tercil 1	-26.26***	-19.30***	-24.58***
	(5.87)	(5.94)	(4.60)
Tercil 2	0.913	4.153	-1.066
	(12.25)	(13.42)	(13.64)
Tercil 3	18.47	20.55	13.26
	(17.89)	(18.07)	(18.09)

Results of the equation **??** for the consumption of soft drinks, expressed in liters per household. SP: Main breadwinner of the household. Clustered standard errors at the Autonomous Community level.*** p<0.01, ** p<0.05, * p<0.1

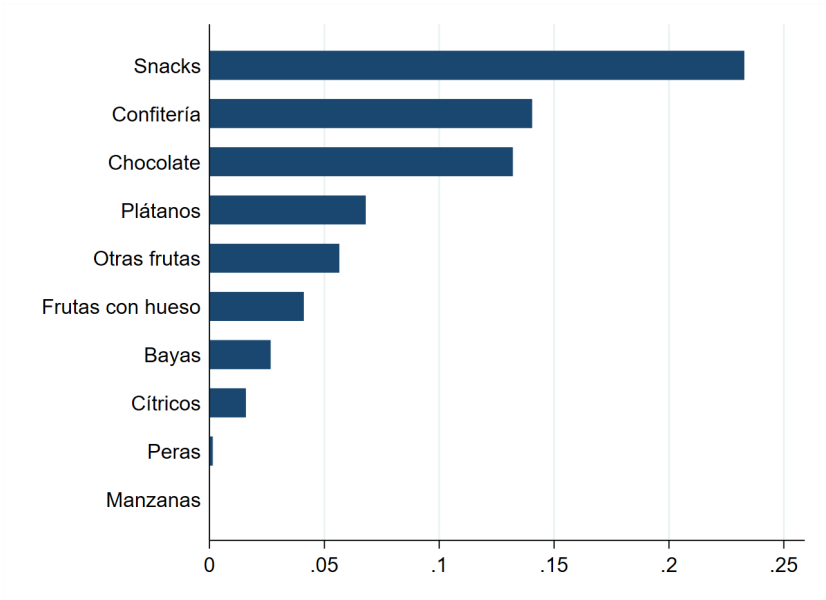
5.3 Effects on Complementary Goods

As discussed in more detail in the literature section, the literature studying this type of tax has focused on analysing the consumption of substitute goods, mostly soft drinks that are not subject to the tax because of their low sugar content. At this point, we have documented that the policy had a significant and negative impact on a portion of the poorest households. This opens the door to another relevant question, whether the decline in consumption of sugar-sweetened and edulcorated beverages also implies a decline in the consumption of complementary goods. This is a question that plays a key role in the ultimate effect of the policy on health and which, so far, has not received sufficient attention in the literature. The main papers that have studied how such taxes affect the consumption of substitute and complementary goods are (Schroeter et al., 2008) and (Finkelstein et al., 2013), theoretically and empirically respectively. The latter find that the fall in the consumption of complementary goods, as a consequence of a tax on sugar-sweetened beverages, contributes significantly to reducing the purchase of calories.

Assuming that complementary goods are also unhealthy products, a tax-induced reduction in their consumption would amplify the final effect of the policy on public

health, without the need to establish new taxes. Thanks to the design of the Household Budget Survey (HBS), we have available household expenditure on a broad set of foods, although we will focus our analysis on expenditure on *snacks* products, following (Edwards, 2011). A technical justification for this decision can be found in the graph below, which shows the correlation of expenditure per household on various goods with expenditure on soft drinks. As can be seen, the correlation of expenditure on soft drinks is particularly high for unhealthy products such as *snacks* (a group mostly composed of crisps), and, to a lesser extent, confectionery, and chocolate. However, the correlation is much weaker, or even zero, for healthy products such as a wide range of fruit.

Figure 7: Correlation of each expenditure with the expenditure per household on soft drinks for the first tercile of equivalent expenditure (2017-2020)

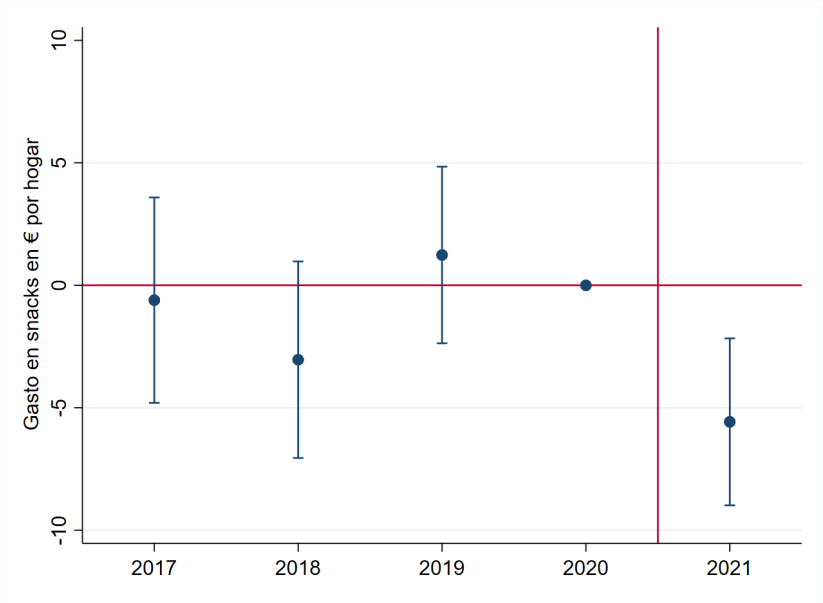


Ideally, we would estimate the equation 1 for the quantity consumed of *snacks*, but, as mentioned in the section 3.1, there are certain expenditure codes for which either the quantity consumed does not exist or has a particularly high level of missing values, the latter being the case for *snacks*. This is why we estimate the equation 1 using the expenditure per household on *snacks*, we consider that, as there are no abrupt price changes between the pre and post periods, the expenditure is a suitable proxy for the quantity consumed which, unfortunately, we cannot observe with the available data.

Before turning to the equation 1 we estimate, for expenditure on *snacks*, the equation 3 to test the key assumption of parallel prior trends. The coefficients associated with the interaction between treatment and pre-policy years are statistically insignifi-

cant at 5%, while the coefficient associated with the policy year is statistically significant and economically relevant, with a fall in expenditure per household of 5 €. With respect to the average expenditure on *snacks* by poor households in the year prior to the implementation of the policy, this implies a drop in expenditure of 10.5%, an amount remarkably similar to the drop in soft drink consumption documented among the same group of households in the previous section.

Figure 8: Effect of the VAT increase in 2021 on spending on *snacks* among households in the first tercile of equivalent spending



In the following table we present the results, for different levels of specification, of the equation 1 for the expenditure per household on *snacks*. Again, as with the regressions for soft drink consumption, the variation in coefficients across specifications is minimal, with our preferred specification yielding a 5€ drop in expenditure per household. The policy relevance of these results can hardly be ignored, since they imply an elasticity of consumption of complementary products with respect to soft drink consumption very close to unity.

This implies that, solely through a contagion effect, households have reconfigured their consumption baskets even in goods that were outside the scope of the tax. This finding highlights the importance of taking into account, when assessing taxes on sugar products, the set of complementary products that surround them and which, until now, have not been studied in depth in the literature. In the appendix we find evidence that the fall in expenditure that we estimate as a consequence of the tax is essentially

explained by a fall in expenditure within the treatment group, and not only by an increase in consumption by households in the control group. This, coupled with the very similar level of pre-policy spending on snacks between the two groups, gives us confidence in the robustness of our results.

Table 6: Effects of the VAT increase in 2021 on spending on *snacks*

Variables	(1) Sin controles	(2) Controles SP	(3) Controles SP y hogar
snacks	-5.194***	-5.411***	-4.966***
	(1.477)	(1.513)	(1.396)
Observations	34,290	34,290	34,290

Results of the equation 1 for spending on *snacks*, expressed in euros per household. SP: Main breadwinner of the household. Clustered standard errors at the Autonomous Community level.*** p<0.01, ** p<0.05, * p<0.1

6 Conclusions

Within the empirical and health debate on obesity, especially childhood obesity, policies around sugar consumption have been gaining weight and have come to occupy the centre stage. This is especially true for taxation, considered within the economic discipline as one of the instruments with the greatest potential to incentivise or disincentivise certain behaviours that generate externalities. However, the methodological challenges in assessing taxes on sugar-sweetened and edulcorated beverages are numerous. First, many of these taxes lack a clear control group that can be used to assess their impact, as they are typically applied across the board to all households in a country. Moreover, the lack of adequate household consumption data means that we often have to rely on retail sales data, which ultimately prevents us from disaggregating the effect of the tax across different types of households.

In this paper, we present quasi-experimental evidence on the effect of a VAT increase from 10% to 21% on sugar-sweetened and edulcorated beverages in Spain. This policy has three peculiarities with respect to other taxes on sugar-sweetened beverages: first, it is an ad valorem tax that does not vary according to the amount of sugar; second, sugar-sweetened beverages are also affected; and finally, beverages consumed in on-

trade establishments were excluded from the tax increase. Although the VAT increase does not only apply to soft drinks, as it also includes juices or energy drinks, we focus on this group because of its larger size in terms of the number of households with positive consumption and the fact that all products in this group, even if they are zero sugar, were affected by the tax.

Our empirical strategy is to use as a control group households in three Spanish regions not affected by the tax, the Canary Islands, Ceuta and Melilla, which have their own indirect taxes. As they are located outside mainland Spain, they do not present any risk of consumer mobility across regions. First, we find clear evidence of a pass-through of the tax to prices of more than 90%, concentrated in the first two months of the tax's entry into force. Secondly, we find that only 33% of households with a lower level of equivalent expenditure show a negative and statistically significant fall in consumption of 13%, essentially via the intensive margin. Moreover, within the first tertile of equivalent expenditure, households with children aged 5-16 have reacted more strongly to the policy, with a five-fold fall in their soft drink consumption, equivalent to 25 litres per household per year. Finally, we make a particularly novel contribution to the literature by estimating the effect of the tax on complementary goods. Specifically, we follow (Finkelstein et al., 2013) studying the impact it had on spending on *snacks*, which has the highest correlation with spending on soft drinks at the household level. We find that, among poor households, spending on *snacks* fell by 11%, a drop almost equivalent to the decline in soft drink consumption previously documented. This finding highlights the importance of the spillover effects of such taxes on complementary goods, because of their potential effects on health variables.

These results are consistent with previous evidence, which shows a high degree of pass-through of such taxes to prices, as well as a limited effect on consumption for price increases of less than 20%. In addition to adding to this growing literature on the effects of such taxes, our key contribution is the disaggregation of the effect across different types of households according to their economic status and household composition. As noted in (Sassi et al., 2018), the identification of these types of households, and ultimately individuals, is key to subsequently assessing the health consequences of the tax. Positive health effects, coupled with a lower share of health spending in the

budgets of poor households, significantly reduce the negative distributional effect that such taxes tend to have. A more complete and richer evaluation of these effects is, in any case, the next logical and necessary step in this paper.

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A Anexo

En el siguiente anexo se presentan diferentes cuadros con datos y resultados a los que se ha hecho referencia a lo largo del texto principal pero que, por razones de espacio, no se han podido incluir en el resto de secciones del documento. En primer lugar, en el cuadro cinco podemos ver los datos a partir de los cuales se construye el primer gráfico de la sección 5 que muestra la evolución del precio medio del litro de refrescos diferenciando entre los hogares que forman parte del grupo de control (Canarias, Ceuta y Melilla) y los hogares tratados (Peninsula y Baleares).

En el cuadro siete se muestran los resultados de la ecuación 1 considerando un grupo de gasto diferente, en este caso el de las bebidas energéticas. Este grupo de bebidas no fue incluido en el análisis principal por el reducido tamaño de su consumo en términos agregados, principalmente motivado por el reducido % de hogares que presentan un consumo positivo de dicho bien.

Table 7: Precio medio por litro de refrescos según su exposición a la subida del IVA

	(1)	(2)
Año	Tratados	Controles
2017	1.100	.938
2018	1.156	.954
2019	1.206	1.007
2020	1.246	1.065
2021	1.386	1.098

Fuente: Elaboración propia a partir de los microdatos de la EPF (2017-2021)

Los resultados indican que el efecto sobre el consumo es igualmente negativo en el primer y segundo tercil. Sin embargo los resultados en el tercer tercil tienen un signo positivo, si bien solo son significativos al 10%. De igual modo que en el análisis principal, los resultados del tercer tercil deben ser tomados con prudencia por la escasa muestra entre el grupo de control. En el cuadro ocho, se muestran los resultados para las bebidas isotónicas con resultados marcadamente similares respecto al primer tercil.

En este caso se encuentra un efecto positivo dentro del segundo tercil aunque no tan significativo como el efecto encontrado en el primero (significativo al 5% pero no

Table 8: Efecto de la subida del IVA en el consumo de bebidas energéticas por terciles

	Sin controles	Controles SP	Controles SP y hogar
Tercil 1	-1.135***	-1.097***	-1.103***
	(0.319)	(0.314)	(0.302)
Tercil 2	-1.861***	-1.979***	-1.977***
	(0.377)	(0.374)	(0.360)
Tercil 3	1.470**	1.249*	1.203*
	(0.618)	(0.626)	(0.631)

Fuente: Elaboración propia a partir de los resultados de la ecuación **??**

Table 9: Efecto de la subida del IVA en el consumo de bebidas isotónicas por terciles

	Sin controles	Controles SP	Controles SP y hogar
Tercil 1	-1.657***	-1.532***	-1.553***
	(0.366)	(0.384)	(0.395)
Tercil 2	1.223*	1.273**	1.354**
	(0.598)	(0.539)	(0.551)
Tercil 3	-0.152	-0.324	-0.314
	(0.518)	(0.449)	(0.414)

Fuente: Elaboración propia a partir de los resultados de la ecuación **??**

al 1%). Todo ello indica que, de incluir estos productos dentro del análisis general, las conclusiones que se han extraído no se verían afectadas.

Por otra parte, la decisión adoptada en el documento principal de presentar los resultados por separado para los refrescos, las bebidas energéticas y las bebidas isotónicas en el apartado del análisis de precio responde a las potenciales diferencias tanto en la estructura de la demanda como en su elasticidad respecto al precio y, en este caso, a los impuestos. Del mismo modo, se ha considerado que los resultados respecto al consumo merecían presentarse en este trabajo.

Table 10: Consumo medio de refrescos por hogar en litros (2017-2021)

	Tercil 1		Tercil 2		Tercil 3	
	Tratados	Controles	Tratados	Controles	Tratados	Controles
2017	101.17	99.11	113.11	98.81	117.55	97.16
2018	90.83	89.03	125.12	94.56	99.19	91.01
2019	82.80	83.42	106.93	90.80	140.16	84.90
2020	82.81	84.21	112.04	96.58	87.78	90.63
2021	74.52	85.43	118.81	96.21	101.96	92.71

Fuente: Elaboración propia a partir de los microdatos de la EPF (2017-2021)

Table 11: Consumo medio de bebidas energéticas por hogar en litros (2017-2021)

	Tercil 1		Tercil 2		Tercil 3	
	Tratados	Controles	Tratados	Controles	Tratados	Controles
2017	2.59	2.81	1.71	4.12	2.55	2.13
2018	1.86	2.95	1.90	3.10	1.19	3.42
2019	2.05	2.43	1.73	2.99	1.23	1.35
2020	1.99	4.19	1.70	2.04	1.70	0.92
2021	1.98	4.97	2.20	5.34	2.05	0.82

Fuente: Elaboración propia a partir de los microdatos de la EPF (2017-2021)

Para finalizar, en las siguientes tablas puede verse el consumo medio por hogar, es decir, considerando también aquellos hogares con un consumo igual a cero, de refrescos, bebidas energéticas y bebidas isotónicas.

Table 12: Consumo medio de bebidas isotónicas por hogar en litros (2017-2021)

	Tercil 1		Tercil 2		Tercil 3	
	Tratados	Controles	Tratados	Controles	Tratados	Controles
2017	3.58	5.34	8.3	5.96	6.18	6.14
2018	2.75	5.12	7.8	5.24	4.2	5.54
2019	3.34	3.61	5.19	5.09	3.41	5.67
2020	5.23	2.94	3.58	4.31	3.34	4.83
2021	4.21	3.09	3.79	3.89	4.46	5.57

Fuente: Elaboración propia a partir de los microdatos de la EPF (2017-2021)

Table 13: Efecto de la subida del IVA en el precio de otros productos

Año	Con controles de hogar y de SP	
	Bebidas energéticas	Bebidas Isotónicas
2017	0.162	-0.357*
	(0.143)	(0.189)
2018	0.355	0.163
	(0.224)	(0.255)
2019	0.201	-0.329
	(0.192)	(0.202)
2020	0	0
	0	0
2021	0.894*	0.379**
	(0.420)	(0.139)

Fuente: Elaboración propia a partir de los resultados de la ecuación 1

Table 14: Contraste de hipótesis de tendencias previas paralelas (Regresiones de consumo en litros del hogar)

Tercil	Sin controles	Controles SP	Controles de SP y hogar
Tercil 1	0.80	1.33	0.51
	0.3825	0.2636	0.4827
Tercil 2	2.58	1.87	3.02
	0.1258	0.1887	0.0991
Tercil 3	3.71	5.32	5.94
	0.0699	0.0331	0.0254

Fuente: Elaboración propia a partir de los resultados de contraste de existencia de tendencias previas paralelas para las regresiones de consumo por hogar en litros
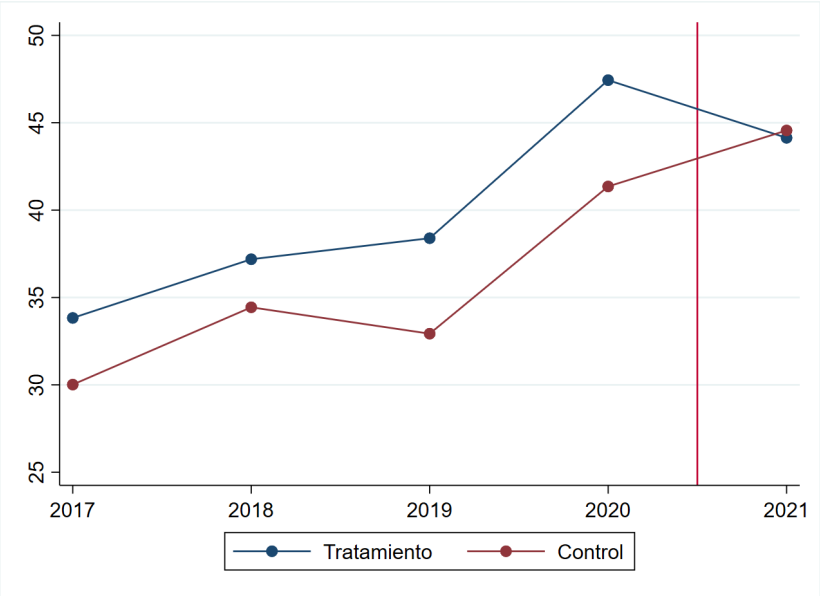
Figure 9: Gasto medio por hogar en *snacks* entre hogares del primer tercil de gasto equivalente en € (2017-2020)